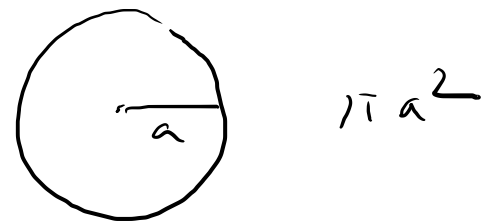


Area.



$$\text{Area} = \iint 1 \, dy \, dx \quad / \quad = \iint 1 \, dx \, dy$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx \, dy = r \, dr \, d\theta$$

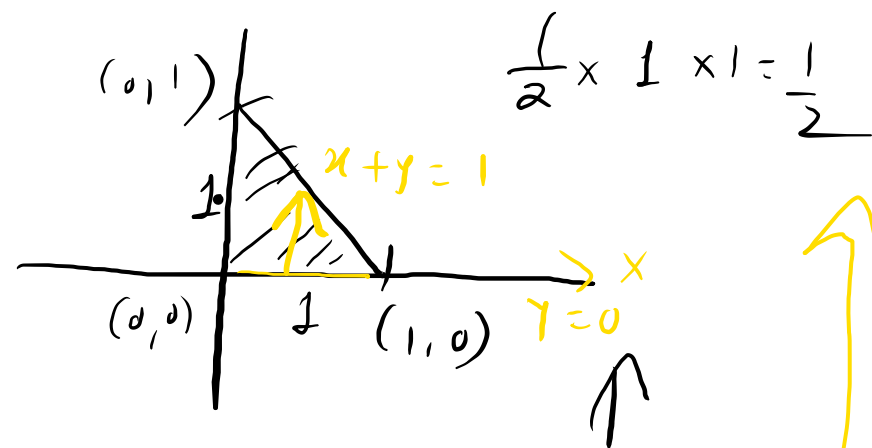
$$\text{Area} = \iint r \, dr \, d\theta$$

$$\int_{x=0}^1 \int_{y=0}^{1-x} dy \, dx$$

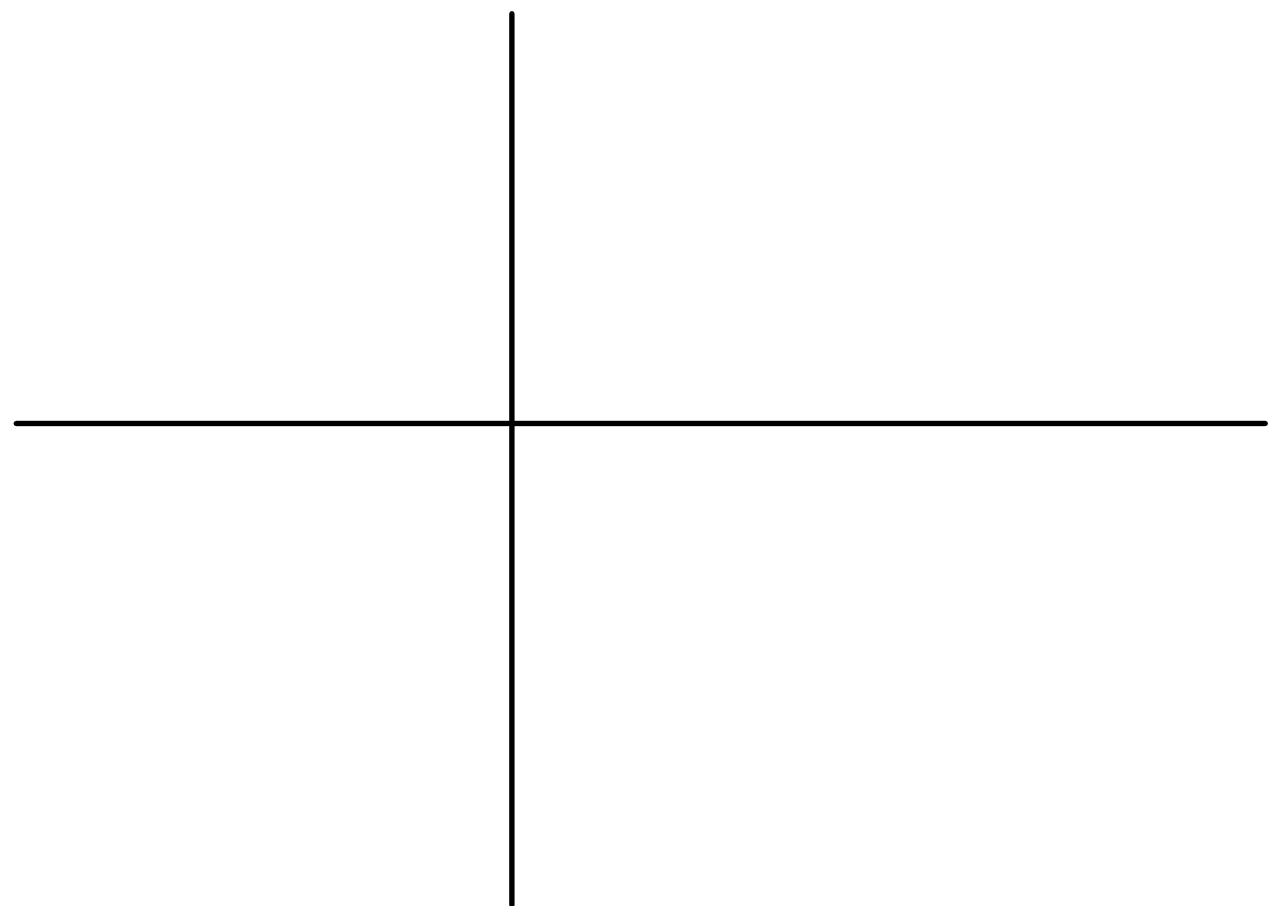
$$= \int_0^1 (y)_0^{1-x} dx$$

$$= \int_0^1 (1-x) dx$$

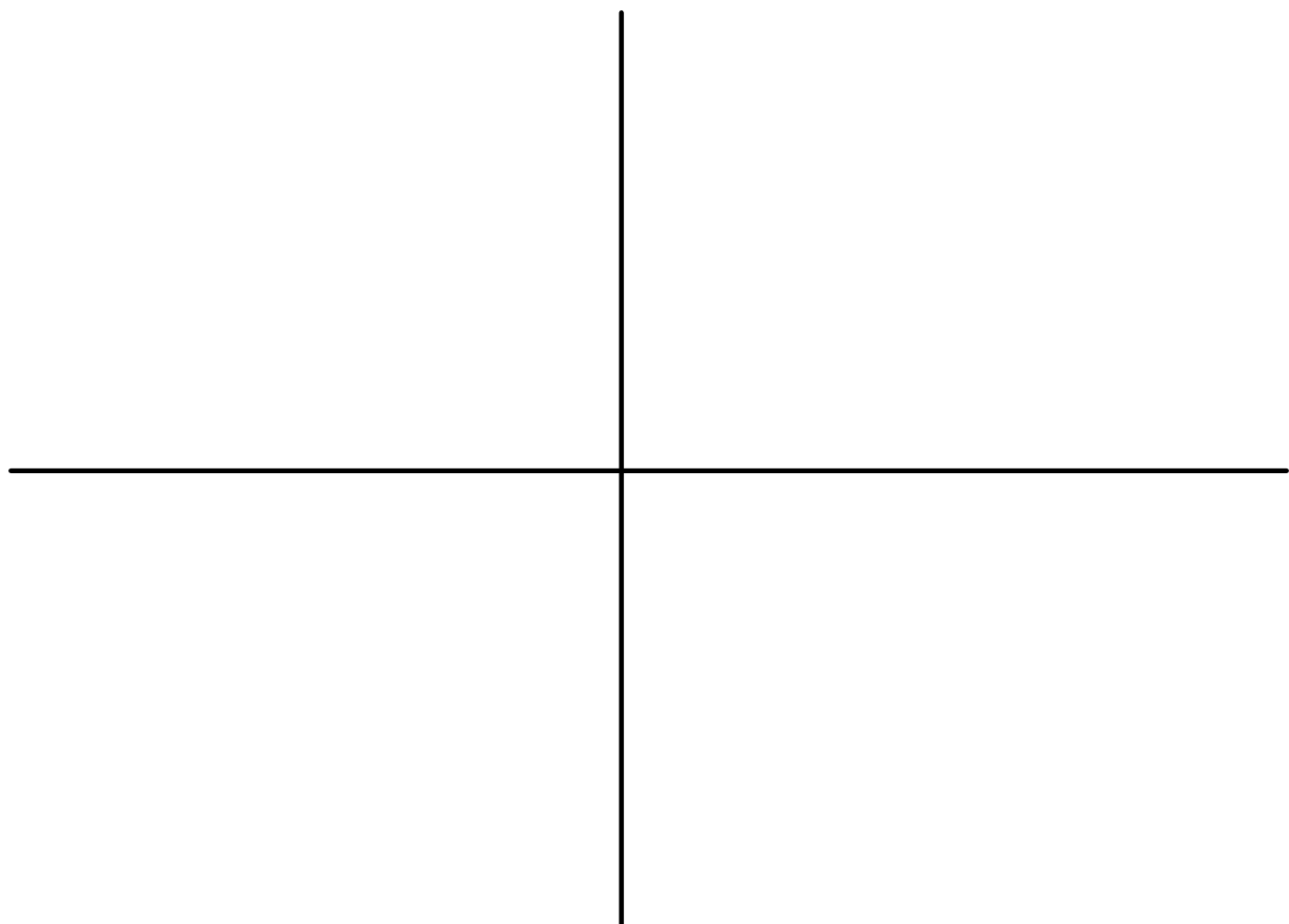
$$= \left[x - \frac{x^2}{2} \right]_0^1 = \left[1 - \frac{1}{2} \right] = \frac{1}{2}$$



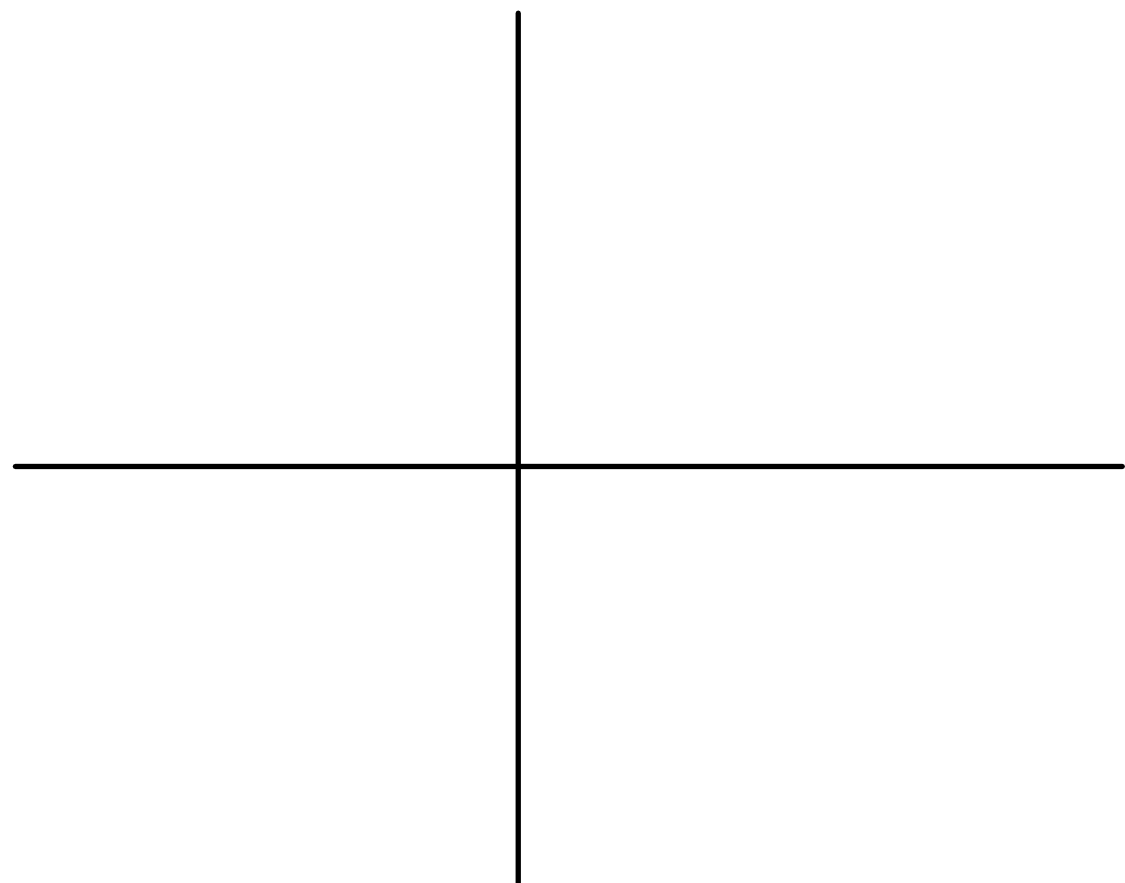
2. Find the area between parabola $y = x^2 - 6x + 3$ and the line $y = 2x - 9$



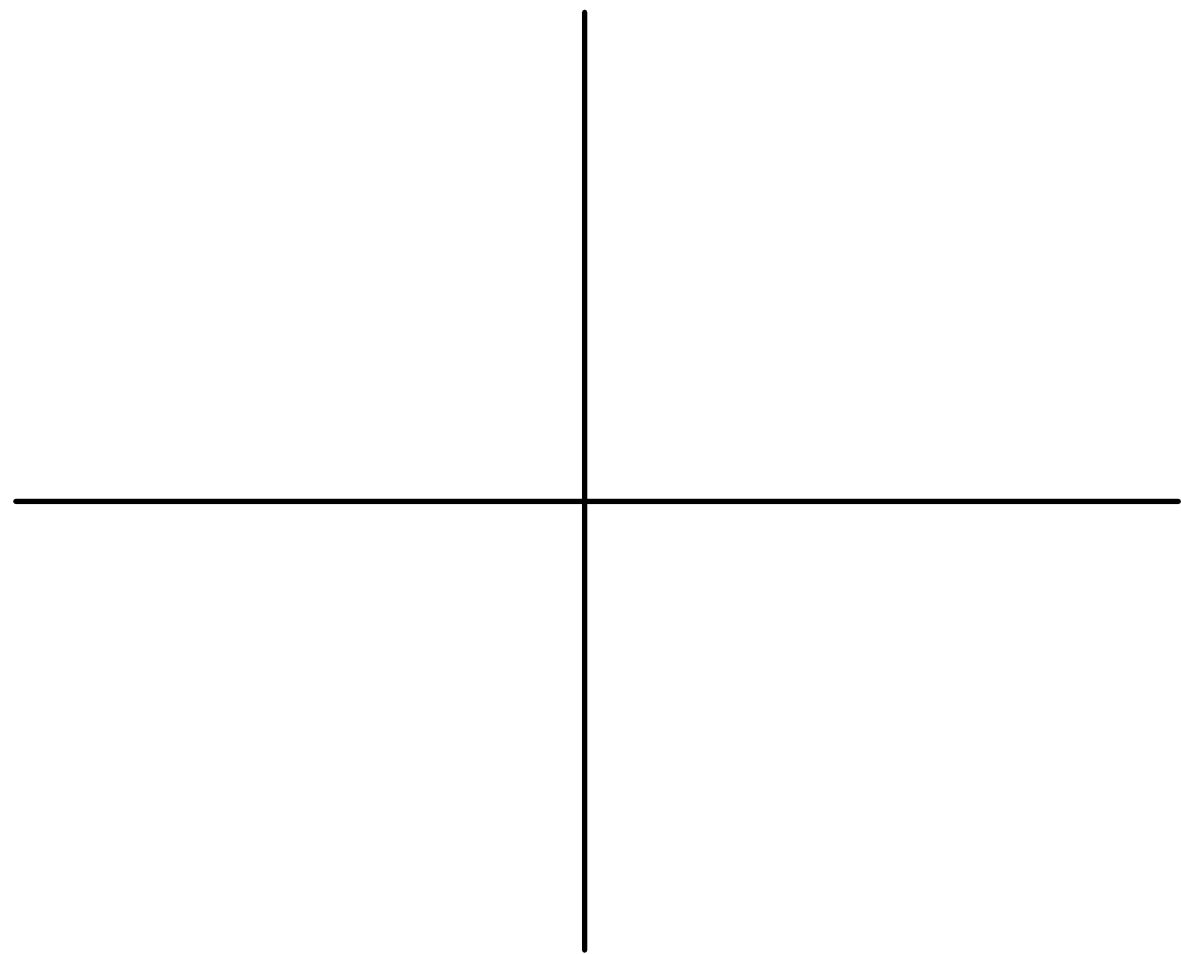
3. Sketch the region bounded by the curves $xy = 16$, $y = x$, $x = 8$ and $y = 0$.
Express the area of this region as a double integral in two ways



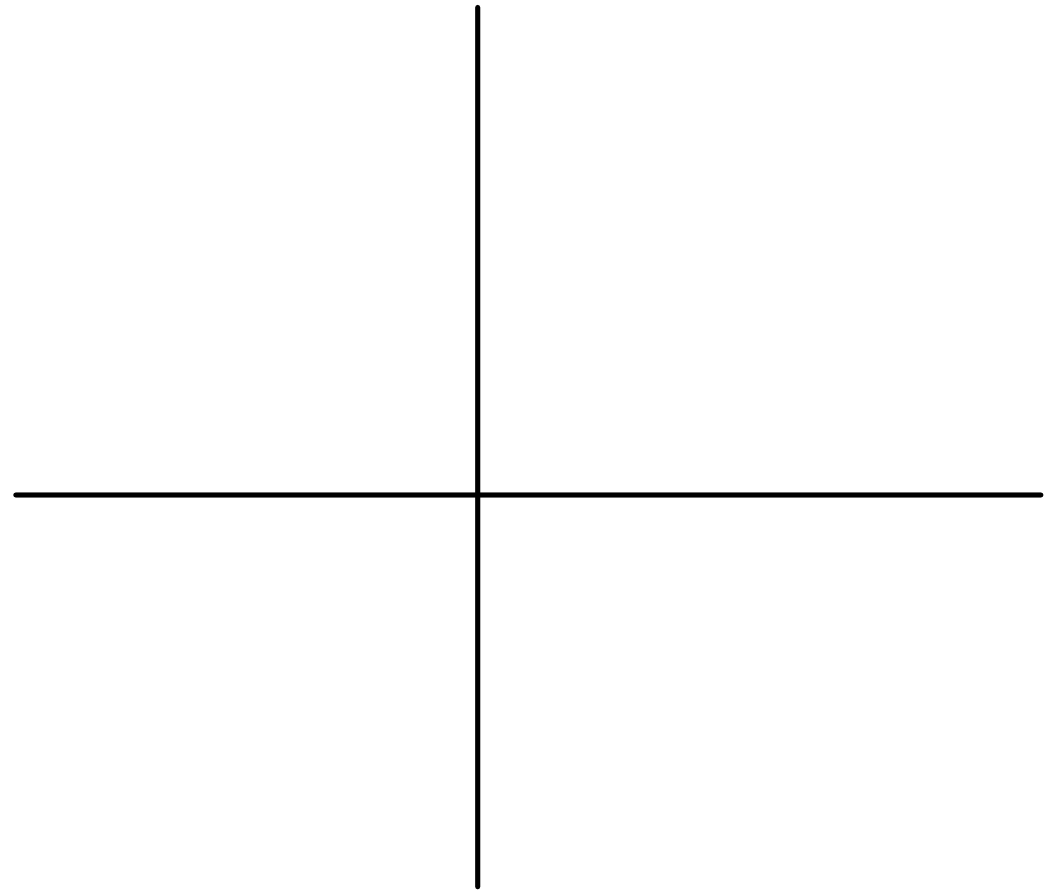
Find by double integration the area of the smaller region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the line $\frac{x}{a} + \frac{y}{b} = 1$



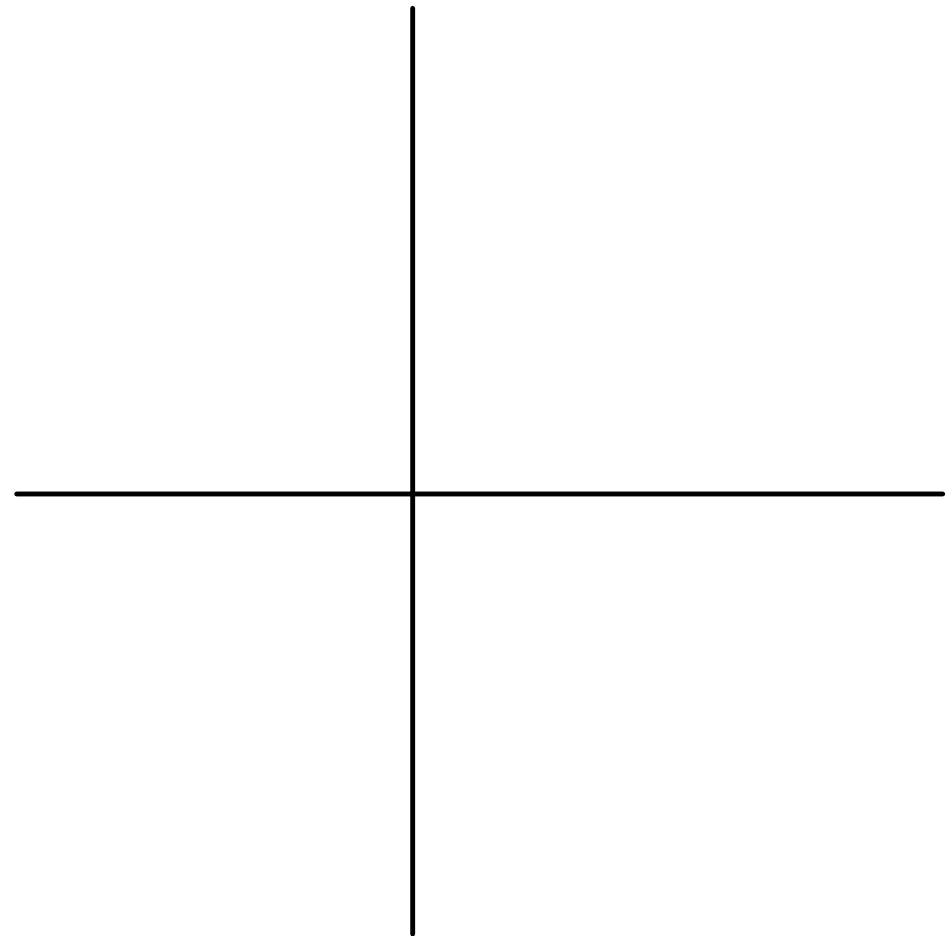
5. Using double integration find the area bounded by the parabolas $x = y^2, x = 2y - y^2$



7. Find the larger of the two areas into which the circle $x^2 + y^2 = 16a^2$ is divided by the parabola $y^2 = 6ax$



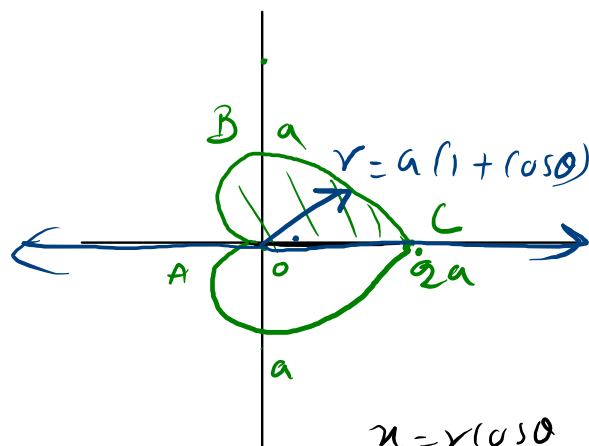
8. Find by double integration the area common to the circles $x^2 + y^2 - 4y = 0$ and $x^2 + y^2 - 4x - 4y + 4 = 0$



9. Find the area of the cardioid $r = a(1 + \cos \theta)$

Solution:

θ	$\cos \theta$	r
0	1	2a
$\pi/2$	0	a
π	-1	0
$3\pi/2$	0	a
2π	1	2a



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$1 + \cos 2\theta = 2 \cos^2 \theta$$

$$1 - \cos 2\theta = 2 \sin^2 \theta$$

$$\text{Area} = 2 (\text{Area ABC})$$

$$= 2 \left(\int_0^{\pi} \int_{r=0}^{a(1+\cos \theta)} r dr d\theta \right)$$

$$= 2 \int_0^{\pi} \left(\frac{r^2}{2} \right)_0^{a(1+\cos \theta)} d\theta$$

$$= \frac{2}{2} \int_0^{\pi} a^2 (1 + \cos \theta)^2 d\theta$$

$$= a^2 \int_0^{\pi} (\cos^2 \theta/2)^2 d\theta = a^2 \int_0^{\pi} (\cos^4 \theta/2) d\theta$$

$$= \text{put } \theta/2 = t \Rightarrow d\theta = 2dt$$

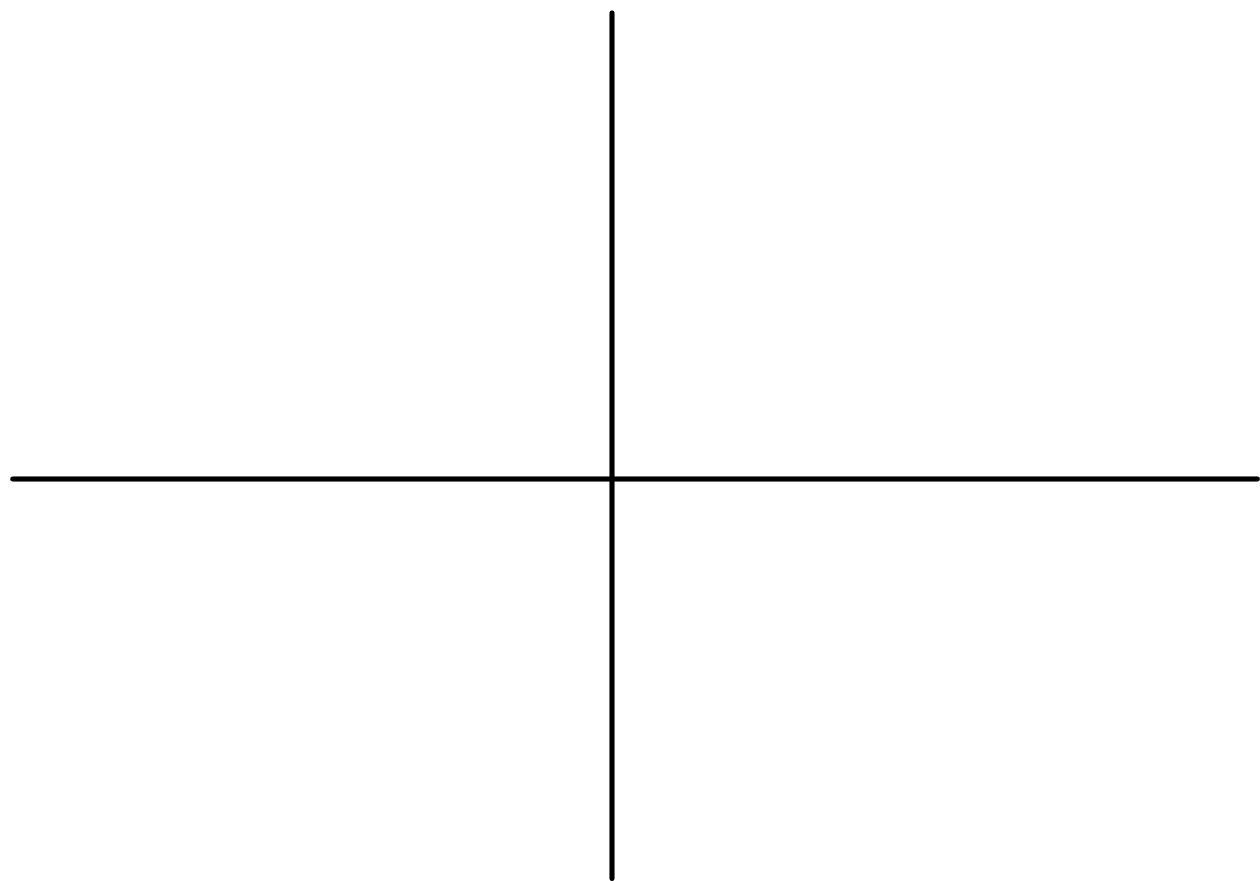
$$\theta: 0 \rightarrow \pi \quad t: 0 \rightarrow \pi/2$$

$$= a^2 \int_0^{\pi/2} \cos^4 t \cdot 2dt$$

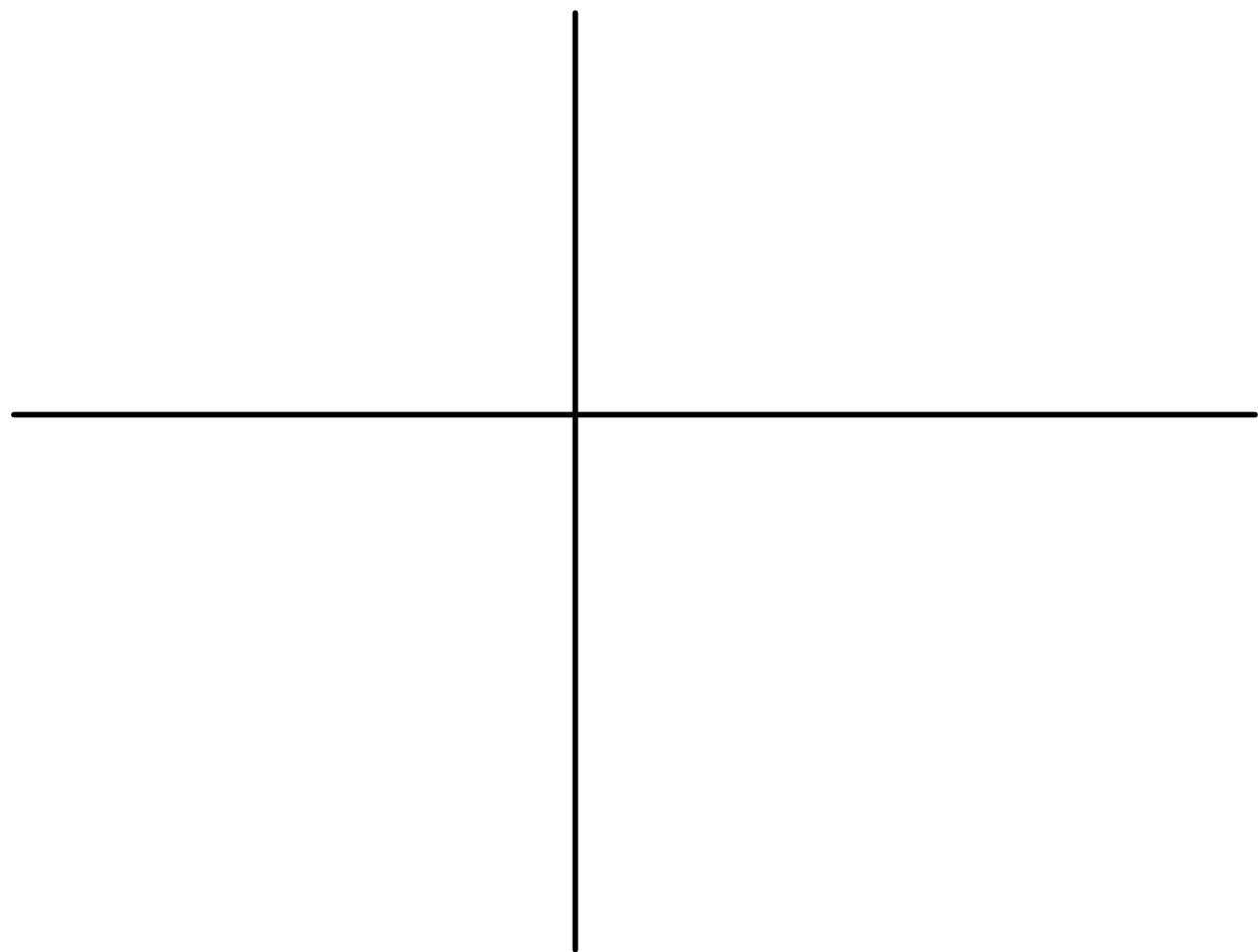
$$= 2a^2 \cdot \frac{1}{2} \beta\left(\frac{0+1}{2}, \frac{4+1}{2}\right)$$

$$= \frac{3}{2} \pi a^2 //$$

- 11.** Find the area inside the circle $r = a \sin \theta$ and outside the cardioid $r = a(1 - \cos \theta)$.

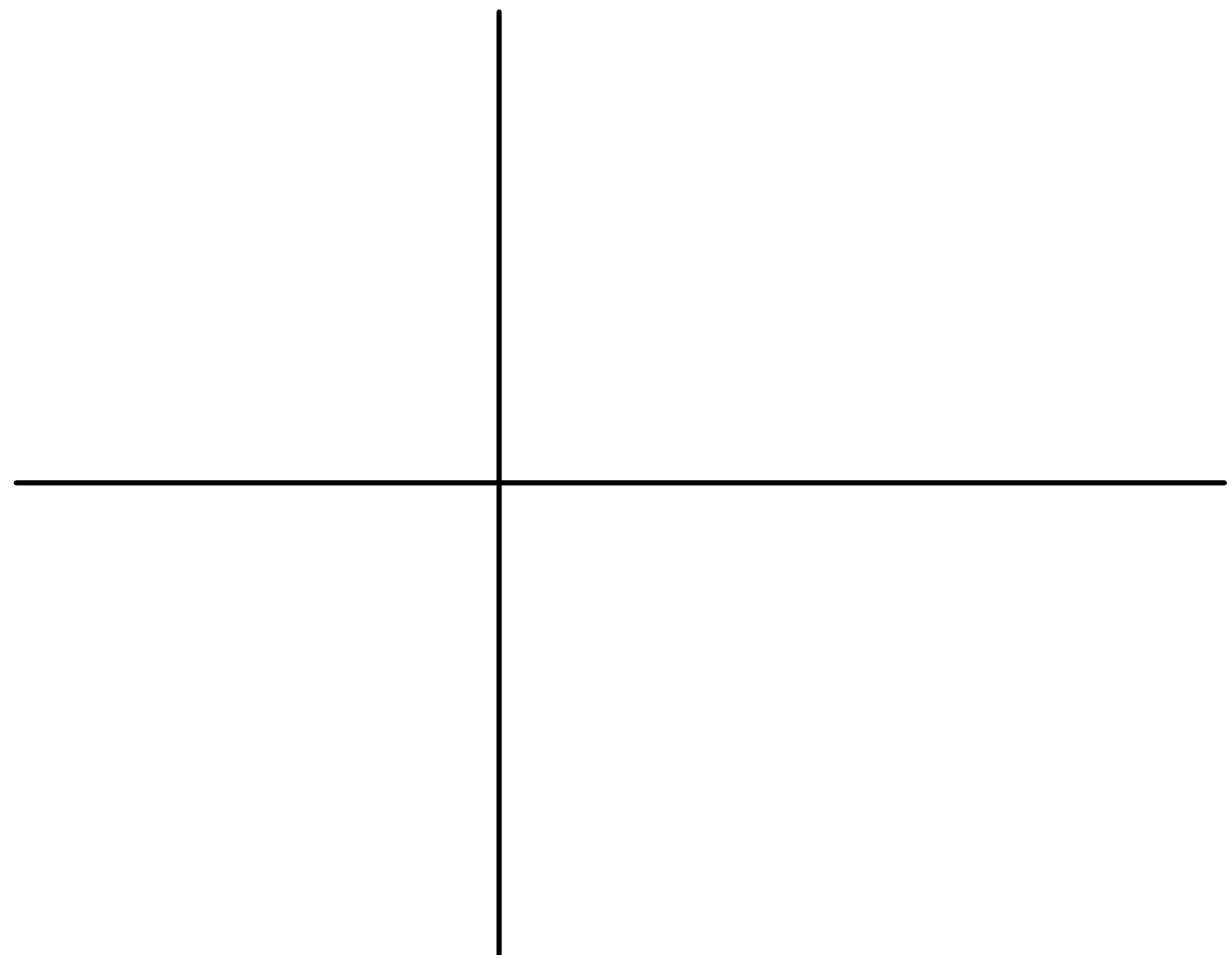


12. Find the area outside the circle $r = a$ and inside the cardioid $r = a(1 + \cos\theta)$.



13. Find the area outside the circle $r = a\sqrt{2}$ and inside circle $r = 2a \cos \theta$.

.....



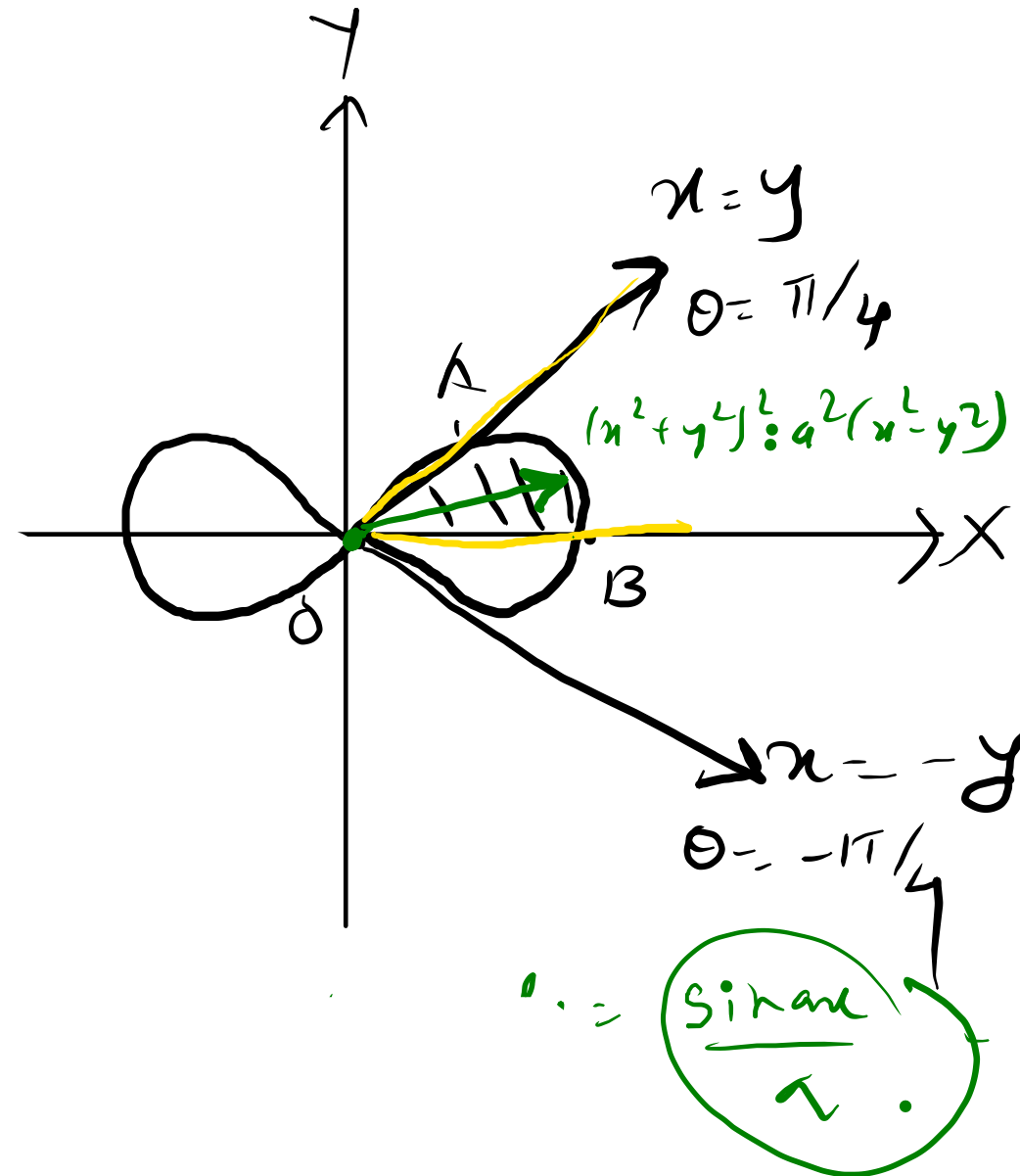
Polar //

10. Find the total area enclosed by the lemniscate of Bernoulli $(x^2 + y^2)^2 = a^2(x^2 - y^2)$

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$dx dy = r dr d\theta$$

$$A = 4 \quad (\text{Area of } OAB)$$



$$r = a \sqrt{\cos 2\theta}$$